

VARDH CHHAJER

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SUMMARY

Computer science student who enjoys building simple, user-friendly software systems. Interested in working with Artificial Intelligence and exploring how technology can be applied to real-world problems. Also curious about Web3, finance, and businesses.

EDUCATION

SVKM's SBMP Polytechnic College

Diploma in Computer Science

India

2023 – Present

TECHNICAL SKILLS

Languages: Python, C#, SQL, TypeScript, Kotlin

Frameworks & Tools: .NET Framework, Visual Studio, Streamlit, React, Vite, FastAPI, Tailwind CSS, Android Studio

Databases: MSSQL, SQLite

Systems: Linux, Docker, WSL, Networking

AI & Blockchain: LLM APIs, Speech-to-Text, Smart Contracts

EXPERIENCE

DeepCytes Cyber Labs

Cyber Analyst

Remote

May 2026 – Jun 2026

- Performed cybersecurity analysis and threat assessment tasks in a fully remote capacity.
- Contributed to cyber intelligence operations and security research within a professional lab environment.

Balar Fabrics Pvt. Ltd.

Software Development and Deployment Intern

Hybrid

Jan 2026 – Present

- **Parts Inventory System:** Built a full-stack Streamlit-based inventory management system with barcode scanning, SQLite backend, and TSPL thermal printer integration for a machinery parts store.
- **ERP Integration:** Implemented bulk data migration pipeline from Excel via Pandas/openpyxl and interactive dashboards with Plotly charts and CSV export.
- **Tech Stack:** Python, Streamlit, SQLite, Pandas, Plotly, pywin32, JavaScript.

Balar Fabrics Pvt. Ltd.

Software Development Intern

On-site

Nov 2024 – Jan 2025

- **Inventory System:** Designed and deployed a barcode-based inventory management system used in production.
- **Tech Stack:** C#, Visual Studio, .NET Framework, Microsoft SQL Server (MSSQL).
- **Networking:** Configured local networking and IP-based deployment in a live environment.

Spectrum'25 College Techfest

Technical Team Member

College Campus

2024

- **Web Development:** Built and maintained the techfest website using React and Node.js.
- **Collaboration:** Provided on-ground technical support and coordinated with teams.

PROJECTS

balar | ERP-to-Cloud Synchronization System

- Engineered a robust SQL Server-to-Cloud sync engine using `pyodbc` and `FastAPI` that performs real-time data extraction of sales orders, invoices, and ledger masters with read-only transaction isolation level settings.
- Implemented a complex financial outstanding calculation engine that replicates `PROC_OUTSTANDING` logic, accurately reconciling bill-wise receivables by aggregating `PARTYDETAIL` bills against `ADJMASTER` receipts and unallocated `TRAN_DETAIL` credits.
- Developed an automated user lifecycle management system that bulk-creates party and agent accounts via `bcrypt` hashing and exports credentials to CSV, ensuring seamless onboarding for ERP stakeholders.

- Designed a resilient data synchronization pipeline that handles multi-year outstanding calculations by connecting to historical databases and applying FIFO credit allocation to ensure accurate financial reporting.
- **Tech Stack:** Dart, Python, C, C++, Swift, JavaScript, FastAPI, SQLAlchemy

Ai Virtual Tryon | *Full-Stack Generative AI Application*

- Engineered a multi-stage garment workflow using a custom React hook that manages state transitions for model uploads, fabric selection, and advanced styling configurations.
- Implemented robust file processing pipelines for PDF and image assets, utilizing `pdf.js` for high-resolution rendering, client-side cropping tools, and automated design number overlay generation.
- Developed a secure authentication system featuring OTP-based verification via Nodemailer and Gmail SMTP, integrated with a time-to-live (TTL) memory store for session management.
- Built a comprehensive usage tracking and cost-analysis engine that calculates token consumption and generation metrics, providing real-time analytics via a dedicated dashboard.
- Designed a prompt-validation layer using keyword-based sanitization and regex-driven input filtering to prevent injection patterns and ensure geometry preservation during AI generation.
- **Tech Stack:** TypeScript, JavaScript, React, Next.js, tailwindcss

Autogit Tool | *Developer Productivity CLI*

- Engineered a recursive project scanner using `readdirSync` and `statSync` to perform deep dependency analysis, identifying build systems, test frameworks, and environment variables across multi-directory architectures.
- Implemented an automated documentation pipeline that integrates with LLM providers to generate context-aware `README.md`, `ARCHITECTURE.md`, and `PROJECT_SUMMARY.md` files based on project-specific code analysis.
- Developed a robust Git automation service using `simple-git` that performs intelligent staging, generates conventional commit messages via diff analysis, and manages remote repository synchronization via the GitHub API.
- Built a cross-platform configuration manager that handles secure API token injection, supports multiple AI providers (OpenAI, Anthropic, Ollama), and automates project publishing and version release workflows.
- **Tech Stack:** TypeScript

Fabric Shortage Calculator Pro | *Category*

- Engineered a fabric shortage calculation algorithm that matches minimum thresholds, calculates longation percentages, and determines balance grey and final shortage values.
- Developed an interactive dashboard using Streamlit, featuring a multi-tab interface, sidebar navigation, and interactive data tables for user input and result visualization.
- Implemented a PDF generation feature using ReportLab, allowing users to export detailed reports with breakdown summaries, calculation traces, and sorted unsorted data.
- Designed and implemented data import and export functionality, enabling users to load and save reports, and persist data across sessions.
- **Tech Stack:** Python, Streamlit, Pandas

Barcode-Based Inventory Management System (Python) | *Industry-Oriented Project*

- Built a Streamlit-based inventory management system with real-time barcode scanning, multi-page navigation, and SQLite backend for a machinery parts store.
- Designed a relational database schema (SQLite) with 6 normalized tables — parties, items, machines, persons, parts inward, and parts usage — with foreign key relationships and computed remaining quantities.
- Developed TSPL thermal printer integration (TSC TTP-244 Pro) via Windows Print Spooler (`pywin32`), generating Code 39 barcodes across a 3-column label grid with sub-millimeter alignment calibration.
- Engineered a global barcode scanner intercept using injected JavaScript that detects rapid keystroke patterns (<500ms) to distinguish scanner input from human typing, auto-redirecting to the usage page with pre-filled data.
- Implemented ERP master data migration pipeline — bulk import of parties and items from Excel (`.xlsx`) via Pandas/openpyxl with full database reset capability.
- Created interactive dashboard with Plotly charts, CSV export, searchable records, and CRUD operations for inward entries and usage records.
- **Tech Stack:** Python, Streamlit, SQLite, Pandas, Plotly, TSPL/Code 39, JavaScript, pywin32, openpyxl.

Barcode-Based Inventory Management System | *Industry-Oriented Project*

- Built a barcode-based inventory system to manage stock movement and material records in a live environment.

- Deployed in production with barcode scanners and local network integration.
- **Tech Stack:** C#, .NET Framework, Visual Studio, MSSQL.

Finsakhi – Financial Education Platform | *Hackathon Project*

- Multi-module application featuring financial education, assessments, investments, podcasts, and AI chatbot support.
- Python-based backend with modular routers for lessons, stories, and podcasts; React + Vite frontend for user engagement.
- **Tech Stack:** Python, React, Vite, FastAPI.

HealthBridge – Integrated Health Management Platform | [healthbridge.page](#)

- Platform for blood report analysis, health tracking, meal planning, and AI-powered chat support.
- Python APIs with modular services for uploads, secure data storage, rate limiting, and AI-driven recommendations.
- **Tech Stack:** Python, React, Vite, Tailwind CSS.

Hides – Secure File Storage Application | [hides.app](#)

- FastAPI backend with modular components for authentication, cryptography, email, and file storage.
- Static frontend with wordlist extraction and cryptographic testing scripts.
- **Tech Stack:** Python, FastAPI, HTML, CSS, JavaScript.

StudyFlow – Project & Document Management App | *Personal Project*

- Web application for managing projects, document uploads, and study workflows with PDF export and database integration.
- Modular UI components, user settings with API key management, and context-driven state management.
- **Tech Stack:** React, TypeScript, Vite.

Secure Messaging Application (Matrix-Based) | *Personal Project*

- Deployed and configured a self-hosted Matrix server with built-in end-to-end encryption.
- **Tech Stack:** Linux, Matrix Server.

PROJECTS IN PROGRESS

SMS Forwarder Pro | *Android Application – Kotlin*

- A lightweight, privacy-first Android SMS forwarder built to solve a real-world business problem: reliably delivering banking OTPs from a registered family business number without manual intervention.
- **Privacy & Security:** Messages are forwarded directly from device to recipient with no third-party servers or intermediaries involved – critical for sensitive banking communications.
- **Reliability:** Runs persistently as a foreground service, remaining fully functional when the app is closed or the screen is off – addressing a core failure point of existing solutions.
- **Multi-Channel Forwarding:** Supports forwarding via SMS, Email, and Telegram, with customizable filters, local message logs, and per-rule configuration.
- Actively expanding with additional forwarding channels and features; designed for extensibility from the ground up.
- **Tech Stack:** Android Studio, Kotlin.

Jarvis – Speech-to-Speech AI Assistant | *Personal Project*

- Speech-based AI assistant with conversational memory and natural voice responses.
- **Tech Stack:** Python, Streamlit, OpenAI Whisper, Google Gemini API.

Community Stablecoin System | *Blockchain Project*

- Developing a community-focused stablecoin using smart contracts on Solana and Algorand.
- **Tech Stack:** Solana, Algorand.

CERTIFICATIONS

Google Cybersecurity Professional Certificate

Google

Online

2023

SQL (Basic) Certificate

HackerRank

Online

2025

SQL (Advanced) Certificate

HackerRank

Online

2025